Pocket SDR - An Open-Source GNSS SDR, ver. 0.21

2026-10-21

Overview

Pocket SDR is an open-source Global Navigation Satellite System (GNSS) receiver based on software-defined radio (SDR) technology. It consists of RF frontend devices named **Pocket SDR FE**, utilities for these devices, and GNSS SDR applications (APs) written in Python, C, and C++. It supports almost all signals for **GPS**, **GLONASS**, **Galileo**, **QZSS**, **BeiDou**, **NavIC**, and **SBAS**.

The Pocket SDR FE device includes 2, 4, or 8 RF frontend channels, supporting the GNSS L1 band (1525 - 1610 MHz) or L2/L5/L6 bands (1160 - 1290 MHz). For these signal bands, refer to **GNSS Signal Bands**. Each RF channel of the Pocket SDR FE device provides IF (intermediate-frequency) bandwidth of up to 36 MHz. The ADC sampling rate can be configured up to 32 Msps (FE 2CH) or 48 Msps (FE 4CH and FE 8CH).

Pocket SDR also includes utility programs to configure the Pocket SDR FE devices, capture, and dump digitized IF data. Additionally, Pocket SDR provides GNSS SDR APs to display the PSD (power spectrum density) of captured IF data, search for GNSS signals, track these signals, decode navigation data, and generate PVT (position, velocity, and time) solutions. These utilities and APs are compatible with **Windows**, **Linux**, **Raspberry Pi OS**, **macOS**, and other environments.

The supported GNSS signals by Pocket SDR are as follows. For details on these signals and Pocket SDR signal IDs, refer to **GNSS Signals**.

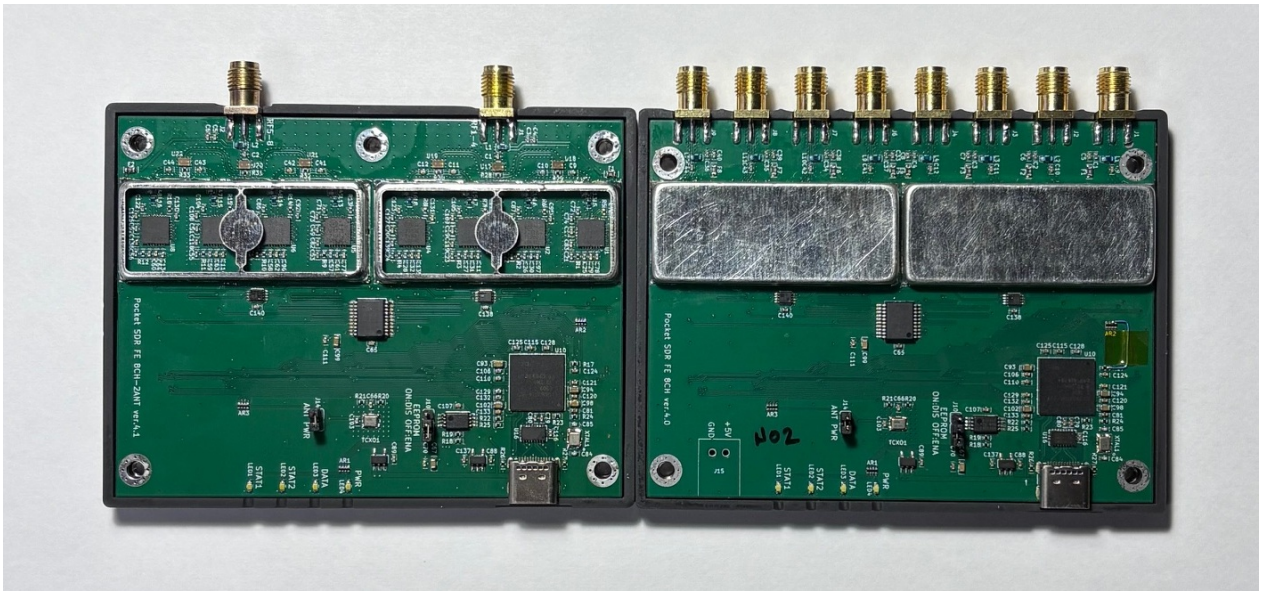
- **GPS**: L1C/A, L1C-D, L1C-P, L2C-M, L5-I, L5-Q
- **GLONASS**: L1C/A (L1OF), L2C/A (L2OF), L1OCd, L1OCp, L2OCp, L3OCd, L3OCp
- **Galileo**: E1-B, E1-C, E5a-I, E5a-Q, E5b-I, E5b-Q, E5AltBOC (E5abQ), E6-B, E6-C
- **QZSS**: L1C/A, L1C/B, L1C-D, L1C-P, L1S, L2C-M, L5-I, L5-Q, L5S-I, L5S-Q, L6D, L6E
- **BeiDou**: B1I, B1C-D, B1C-P, B2a-D, B2a-P, B2I, B2b-I, B3I
- **NavIC**: L1-SPS-D, L1-SPS-P, L5-SPS
- **SBAS**: L1C/A, L5-I, L5-Q

These utilities and APs are written in Python, C, and C++ in a compact and modular way, making them easy to modify for adding custom algorithms.

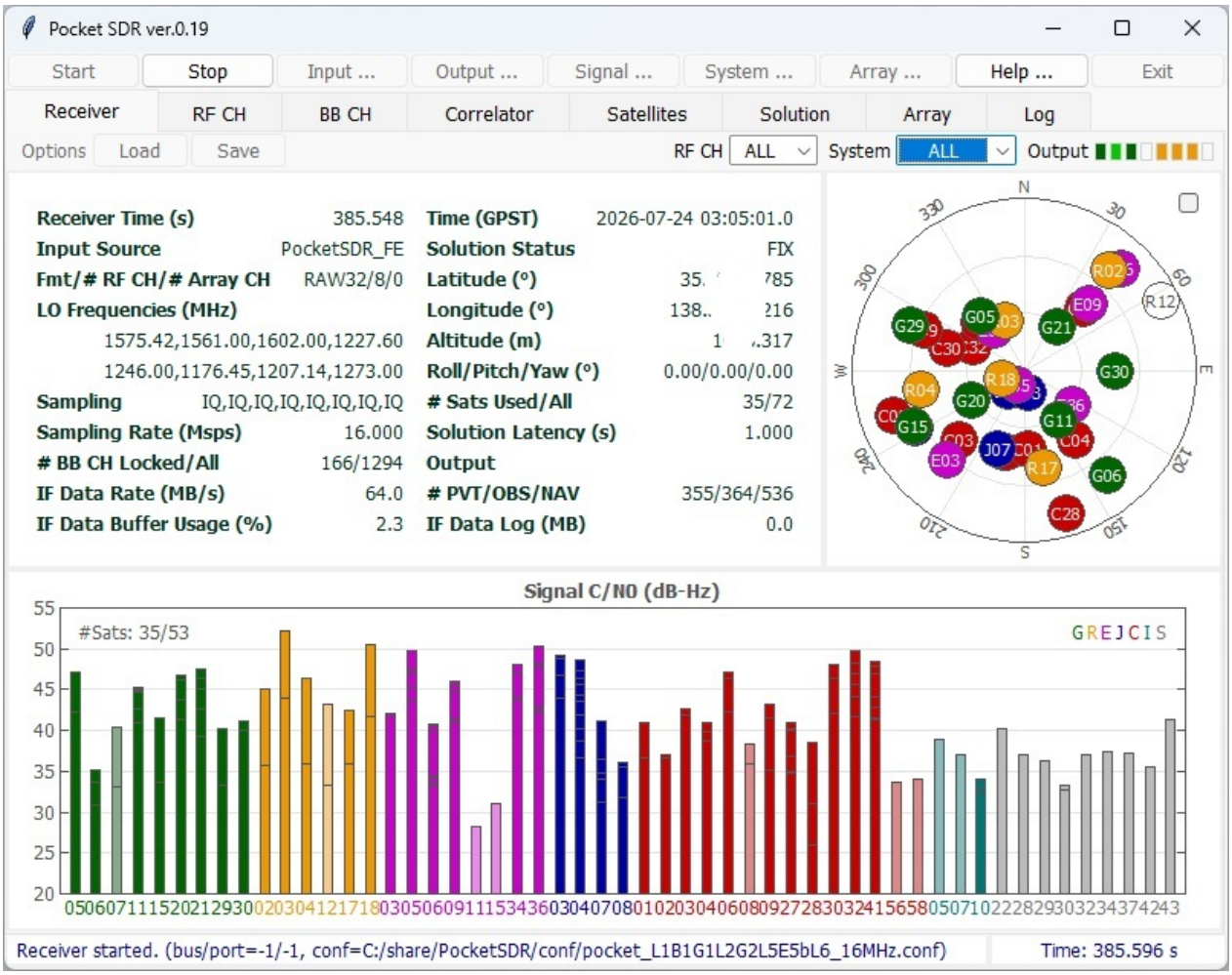
In addition to the Pocket SDR FE devices, Pocket SDR supports third-party SDR hardware (**USRP**, **LimeSDR**, **bladRF**, **PlutoSDR**, **RTL-SDR**, and so on) via **SoapySDR**. See **SoapySDR Device Notes** for verified devices, installation notes, and device-specific caveats.



Pocket SDR FE 2CH (v.2.0 and v.2.3), FE 4CH



Pocket SDR FE 8CH (2-RF-inputs and 8-RF-inputs)



GUI-based Real-Time GNSS SDR Receiver AP

Documents

- Pocket SDR C Library API Reference
  (**https://github.com/tomojitakasu/PocketSDR/tree/master/doc/api_ref.pdf**)
- Pocket SDR Command Reference
  (**https://github.com/tomojitakasu/PocketSDR/tree/master/doc/command_ref.pdf**)
- Pocket SDR GNSS SDR Algorithm Description
  (**https://github.com/tomojitakasu/PocketSDR/tree/master/doc/algo_desc.pdf**)
- Pocket SDR GNSS SDR アルゴリズム解説
  (**https://github.com/tomojitakasu/PocketSDR/tree/master/doc/algo_desc_jp.pdf**)
- T.Takasu, An Open Source GNSS SDR: Development and Application, IPNTJ Next GNSS Technology WG, February 21, 2022 (**https://gpspp.sakura.ne.jp/paper2005/IPNTJ_NEXTWG_202202.pdf**)
- T.Takasu, Development of QZSS L6 Receiver without Pilot Signal by using SDR, IPNTJ Annual Conference, June 10, 2022 (**https://gpspp.sakura.ne.jp/paper2005/IPNTJ_20220610.pdf**)
- T.Takasu, Pocket SDR: Design, Implementation and Applications, A seminar for GNSS Software Defined Receivers, Nov 19, 2024
  (**https://gpspp.sakura.ne.jp/paper2005/pocketsdr_seminar_202411_revA.pdf**)

Directory Structure and Contents

```
PocketSDR
├── bin         # Pocket SDR APs binary programs
├── app         # Pocket SDR APs source programs
│   ├── pocket_conf  # Pocket SDR FE device configurator
│   ├── pocket_dump  # Dump digital IF data of Pocket SDR FE device
│   ├── pocket_scan  # Scan and list USB devices
│   ├── pocket_acq   # GNSS signal acquisition
│   ├── pocket_trk   # GNSS signal tracking and PVT generation
│   ├── pocket_web   # GNSS receiver server with Web UI
│   ├── pocket_snap  # Snapshot Positioning
│   ├── pocket_calib # Antenna array attitude / per-CH bias calibration
│   ├── convbin      # RINEX converter supporting Pocket SDR
│   └── str2str      # Stream converter (RTKLIB-derived)
├── src         # Pocket SDR library source programs
├── html        # Web UI files served by pocket_web (JS + CSS, no build step)
├── python      # Pocket SDR Python scripts (incl. pocket_sdr.py GUI)
├── lib         # Libraries for APs and Python scripts
│   ├── win32        # Built libraries for Windows (UCRT64)
│   ├── macos        # Built libraries for macOS
│   ├── linux        # Built libraries for Linux or Raspberry Pi OS
│   ├── build        # Makefiles to build libraries (UCRT64 / Linux / macOS)
│   ├── cyusb        # Cypress EZ-USB API (CyAPI.a) and includes
│   ├── RTKLIB       # RTKLIB source programs based on 2.4.3 b34
│   ├── pocketfft    # PocketFFT (header-only FFT, used internally)
│   └── SoapySDR     # SoapySDR headers + import lib for SDR device support
├── conf        # Configuration files for Pocket SDR FE
├── FE_2CH      # Pocket SDR FE 2CH H/W and F/W
├── FE_4CH      # Pocket SDR FE 4CH H/W and F/W
├── FE_8CH      # Pocket SDR FE 8CH H/W and F/W (3-bit RF samples)
├── driver      # Driver installation instruction for Pocket SDR FE
├── doc         # Documents (incl. api_ref.md, command_ref.md)
├── image       # Image files for documents
├── sample      # Sample digital IF data captured by Pocket SDR FE
└── test        # Test codes, scripts and data
    ├── python       # Python scripts for tests
    ├── script       # Shell scripts for tests
    ├── src          # source programs for tests
    └── utest        # makefile for unit tests
```

Installation for Windows

- Download and extract PocketSDR.zip or clone the git repository **here** to an appropriate directory <install_dir>. to an appropriate directory <install_dir>.

```
> unzip PocketSDR.zip
or
> git clone https://github.com/tomojitakasu/PocketSDR
```

- Install USB device driver for Pocket SDR FE according to **driver/readme.txt** .
- Install **Python** with checking "Add python.exe to PATH".
- Install additional python packages as follows.

```
> pip install numpy scipy matplotlib
```

- (Optional) To use SoapySDR-supported devices (LimeSDR, HackRF, RTL-SDR, bladeRF, etc.), install **radioconda** and add `<radioconda_install_dir>\Library\bin` to PATH. See **SoapySDR Device Notes** for per-device packages, Zadig driver setup, and bladeRF version caveats.
- Add the Pocket SDR binary programs path (<install_dir>\PocketSDR\bin) to the command search path (Path) of Windows environment variables.
- Add the Pocket SDR Python scripts path (<install_dir>\PocketSDR\python) to the command search path (Path) of Windows environment variables.

To rebuild binary programs for Windows

- If you want to rebuild the binary utilities, APs, or shared libraries for the python APs for Windows, you need **MSYS2** with the **UCRT64** environment. The UCRT64 toolchain links against the same C runtime (`ucrtbase.dll`) as **radioconda**'s SoapySDR DLLs, which gives stable high-rate streaming with SDR hardware (e.g., LimeSDR at 64 Msps). The legacy **MINGW64** environment also builds, but suffers significant performance loss with SoapySDR devices due to msvcrt/UCRT mixing.
- Open the **MSYS2 UCRT64** shell and install the development tools and the Python stack used by the python APs:

```
$ pacman -Syy
$ pacman -S git make
$ pacman -S mingw-w64-ucrt-x86_64-gcc
$ pacman -S mingw-w64-ucrt-x86_64-binutils
$ pacman -S mingw-w64-ucrt-x86_64-python
$ pacman -S mingw-w64-ucrt-x86_64-python-numpy
$ pacman -S mingw-w64-ucrt-x86_64-python-scipy
$ pacman -S mingw-w64-ucrt-x86_64-python-matplotlib
```

- If you intend to use SoapySDR-supported devices, install radioconda first. The default radioconda location used by the Makefiles is `/c/Users/<user>/radioconda/Library`. If installed elsewhere, override with `SOAPY_ROOT` on the make command line. See **SoapySDR Device Notes** for details.
- If SoapySDR device support is not required, build `libsdr`, `pocket_trk`, and `pocket_dump` with `USE_SOAPY=0`. Pocket SDR FE USB devices and IF-file based processing still work; only SoapySDR device input is disabled. If you are rebuilding an existing tree, run `make clean` first.

```
$ cd <install_dir>/lib/build
$ make clean && make USE_SOAPY=0 && make install
$ cd <install_dir>/app/pocket_trk
$ make clean && make USE_SOAPY=0 && make install
$ cd <install_dir>/app/pocket_dump
$ make clean && make USE_SOAPY=0 && make install
```

- Move to the library build directory and build libraries.

```
$ cd <install_dir>/lib/build
$ make
$ make install
```

The default radioconda location is `/c/Users/<user>/radioconda/Library`. If installed elsewhere, override on the command line:

```
$ make SOAPY_ROOT=/c/path/to/radioconda/Library
```

- Move to the application program directory and build utilities and APs.

```
$ cd <install_dir>/app
$ make
$ make install
```

- To run unit tests for the libraries:

```
$ cd <install_dir>/test/utest
$ make
$ make test
==> test_sdr_array.exe
test_sdr_array_new_free_api ... OK
test_sdr_array_ant_pos_api ... OK
t...
test_sdr_usb_req_control ... OK
```

Installation for Linux or Raspberry Pi OS

> **Note**: Pocket SDR has been verified on Ubuntu / Debian-based Linux distributions and on **Raspberry Pi OS (Bookworm) on Raspberry Pi 5** (aarch64). The Pi 5 has been confirmed to handle Pocket SDR FE and the apt-installed SoapySDR devices for typical L1 / L2 single-channel use; see the **Notes for Raspberry Pi 5** subsection below for hardware caveats (USB power, USB 3.0 routing, CPU limits).

- You need fundamental development packages and some libraries. Confirm the following packages installed: git, gcc, g++, make, libusb-1.0-0-dev, libfftw3-dev, python3, python3-numpy, python3-scipy, python3-matplotlib, python3-tk
- (Optional) To use SoapySDR-supported devices (LimeSDR, HackRF, RTL-SDR, SDRPlay, bladeRF, etc.), install SoapySDR and the per-device modules:

```
$ sudo apt install libsoapysdr-dev soapysdr-tools soapysdr-module-all
```

See **SoapySDR Device Notes** for per-device package names, udev access, UHD image installation, and the Ubuntu 24.04 bladeRF module update workaround.

- Download and extract PocketSDR.zip or clone the git repository to an appropriate directory <install_dir>.

```
$ unzip PocketSDR.zip
or
$ git clone https://github.com/tomojitakasu/PocketSDR
```

- Move to the library build directory and build libraries.

```
$ cd <install_dir>/lib/build
$ make
$ make install
```

- Move to the application program directory and build utilities and APs.

```
$ cd <install_dir>/app
$ make
$ make install
```

- Add the Pocket SDR binary programs path (<install_dir>/PocketSDR/bin) to the command search path.
- (Recommended) Install the udev rule so the Pocket SDR FE device is accessible to a normal user account (no `sudo` required). The repository ships a ready-to-use rule at `driver/99-pocket-sdr.rules` that grants the `plugdev` group read/write access to Pocket SDR FE 2CH (Cypress EZ-USB FX2LP, VID `04B4` / PID `1004`) and FE 4CH / 8CH (Cypress EZ-USB FX3, VID `04B4` / PID `00F1`):

```
$ sudo cp <install_dir>/driver/99-pocket-sdr.rules /etc/udev/rules.d/
$ sudo udevadm control --reload-rules && sudo udevadm trigger
$ sudo usermod -aG plugdev $USER     # add yourself to plugdev
```

Log out and back in (or reboot) so the new group membership takes effect, then unplug and re-plug the device. After that, `pocket_scan`, `pocket_conf`, `pocket_dump`, `pocket_trk`, and `pocket_sdr.py` all run without `sudo`.

Verify the rule is in effect:

```
$ lsusb -d 04b4:                          # confirm the device is enumerated
$ ls -l /dev/bus/usb/<bus>/<dev>           # group should be "plugdev", mode 0660
$ pocket_scan                              # should list the device
```

If your distribution uses a different group than `plugdev` (e.g. `dialout` on some systems, or systemd-logind's `uaccess` is sufficient on its own), adjust `GROUP=` in the rule file or rely on the `TAG+="uaccess"` line which grants access to the currently logged-in seat user on systemd-based systems.

- If you skip the udev rule, you must run the Pocket SDR utilities under `sudo` to access the USB device:

```
$ sudo pocket_conf ../conf/pocket_L1L6_12MHz.conf
$ sudo pocket_dump -t 10 ch1.bin ch2.bin
$ sudo pocket_sdr.py
```

Notes for Raspberry Pi 5

Raspberry Pi 5 (Cortex-A76 4-core @ 2.4 GHz, aarch64) runs the same build path as desktop Linux above. The Makefiles auto-detect aarch64 and enable NEON SIMD optimizations (`-DNEON`). A few Pi-specific constraints to be aware of:

- **USB power budget**: Pi 5 advertises 5V / 5A on the official PSU but the USB ports share a ~1.6 A budget. USRP B210 (~2 A peak), Pocket SDR FE 8CH, and BladeRF 2.0 typically need either a USB Y-cable (separate +5V feed) or a **self-powered (externally-fed) USB hub**. Symptoms of insufficient power are intermittent USB resets, FPGA bitstream upload failures (USRP), and dropped samples.
- **USB 3.0 routing**: Pi 5 has 2× USB 3.0 + 2× USB 2.0 ports. Always connect high-rate SDRs (LimeSDR, USRP B210, Pocket SDR FE 8CH) to a USB 3.0 port. Verify enumeration speed with:

```
$ lsusb -t
```

The relevant device should show `5000M` (USB 3.0) ? `480M` means the device fell back to USB 2.0, which is bandwidth-limited and will drop samples at high rates.

- **CPU / sample rate limits**: Pi 5 holds up well thanks to NEON SIMD acceleration. Measured load (`pocket_trk` realtime, single-band L1 tracking, all-in-view PRNs):

Device	Sample rate × tracked channels	CPU load (of 400% = 4 cores)
LimeSDR USB	32 Msps × 7 ch	~280%
USRP B210	32 Msps × 8 ch	~290%
PlutoSDR	6 Msps × 10 ch	~121%
RTL-SDR	2.4 Msps × 16 ch	~102%

  32 Msps with many tracked channels is workable but leaves limited headroom for additional bands or PVT processing. **8 Msps** is a comfortable sweet spot for general L1 use on Pi 5. Heavier configs (LimeSDR 60 Msps × 2-band, Pocket SDR FE 8CH × multi-band) will saturate the Pi 5 CPU and are intended for desktop use.
- **PocketFFT vs FFTW3**: PocketFFT (default) works on aarch64 with NEON. If FFT becomes a bottleneck for your workload, FFTW3 is marginally faster on large transforms ? install with `sudo apt install libfftw3-dev` and switch the build per the **Using FFTW3 instead of PocketFFT** section below.
- **Storage**: For long IF data captures (`pocket_dump -t <long>`, `pocket_trk -raw`), use a USB 3.0 SSD or NVMe HAT. The microSD card on stock Pi 5 cannot sustain >50 MB/s writes for extended periods and will drop samples.

Installation for macOS

> **Note**: Verified on Apple Silicon (arm64). Pocket SDR FE 2/4/8CH (USB) has been confirmed working. For SoapySDR devices on macOS, see **SoapySDR Device Notes**.

- You need **Homebrew** as a base package manager. Install Homebrew by following the instructions on its site, then install development tools:

```
$ brew install libusb fftw
```

- (Optional) Install **radioconda** to use SoapySDR devices and a self-contained Python environment:

```
$ curl -L -O
https://github.com/ryanvolz/radioconda/releases/latest/download/Radioconda-
MacOSX-arm64.sh
$ bash Radioconda-MacOSX-arm64.sh
```

For Intel mac use `Radioconda-MacOSX-x86_64.sh`. Known macOS SoapySDR packaging workarounds are collected in **SoapySDR Device Notes**.

- Download and extract PocketSDR.zip or clone the git repository to an appropriate directory <install_dir>.

```
$ unzip PocketSDR.zip
or
$ git clone https://github.com/tomojitakasu/PocketSDR
```

- Build libraries and applications. The Makefiles auto-detect Apple Silicon (`Darwin arm64`) and use `clang++` for C++ sources and pull Homebrew's `libusb` / `fftw` headers from `/opt/homebrew/include`:

```
$ cd <install_dir>/lib/build
$ make
$ make install
$ cd <install_dir>/app
$ make
$ make install
```

- Add `<install_dir>/PocketSDR/bin` to the command search path.
- For SoapySDR-supported devices, use the driver shortcuts in `<install_dir>/app/pocket_trk/pocket_trk.sh`. See **SoapySDR Device Notes**.
- Pocket SDR FE 2CH / 4CH / 8CH USB devices work on macOS via libusb-1.0 with **no driver install needed** (Apple's IOUSB stack provides raw USB access through libusb). Plug in the device and run:

```
$ pocket_scan                                          # device should be listed
$ pocket_conf <install_dir>/conf/pocket_L1L6_12MHz.conf
$ pocket_dump -t 10 ch1.bin ch2.bin
```

Using FFTW3 instead of PocketFFT (optional)

Starting from version 0.15b, Pocket SDR ships with **PocketFFT** (BSD-3-Clause) as the default FFT backend, replacing **FFTW3** (GPL). The change is licensing-driven; for most workloads PocketFFT is competitive, but FFTW3 is still slightly faster on large transforms and supports runtime wisdom (auto-tuned plans) ? `pocket_trk` / `pocket_snap` / `pocket_acq` can read `python/fftw_wisdom.txt` for further speed-up.

To build with FFTW3 instead, install the library and build `libsdr` and the FFT-using applications with `USE_FFTW=1`.

- Install FFTW3 (single-precision):

```
$ pacman -S mingw-w64-ucrt-x86_64-fftw    # MSYS2 UCRT64 (Windows)
$ sudo apt install libfftw3-dev           # Ubuntu / Debian / Raspberry Pi OS
$ brew install fftw                       # macOS
```

- Rebuild from scratch:

```
$ cd <install_dir>/lib/build
$ make clean && make USE_FFTW=1 && make USE_FFTW=1 install
$ cd <install_dir>/app/pocket_acq
$ make clean && make USE_FFTW=1 && make install
$ cd <install_dir>/app/pocket_snap
$ make clean && make USE_FFTW=1 && make install
$ cd <install_dir>/app/pocket_calib
$ make clean && make USE_FFTW=1 && make install
$ cd <install_dir>/app/pocket_dump
$ make clean && make USE_FFTW=1 && make install
$ cd <install_dir>/app/pocket_trk
$ make clean && make USE_FFTW=1 && make install
```

- (Optional, FFTW3 only) Generate a wisdom file tuned for the host CPU. Once present, `pocket_trk`, `pocket_snap`, and `pocket_acq` pick it up automatically.

```
$ cd <install_dir>/app/pocket_trk
$ ./fftw_wisdom -n 4096
$ cp fftw_wisdom.txt ../../python/
```

Note: FFTW3 is GPL-licensed, so a binary linked against it inherits GPL terms. Keep this in mind when redistributing.

Using AVX-512 instructions (optional)

On x86-64 CPUs with AVX-512 (AMD Zen 4 or later such as Ryzen 9000 series, Intel Ice Lake or later), `libsdr` can be built with `USE_AVX512=1` to use the AVX-512 versions of the carrier mixer, correlator, LPF and array covariance kernels (`-DAVX512`) instead of the default AVX2 ones. The APs link `libsdr` statically, so rebuild them after the library. A binary built with `USE_AVX512=1` does not run on a CPU without AVX-512.

```
$ cd <install_dir>/lib/build
$ make clean && make USE_AVX512=1 && make install
$ cd <install_dir>/app
$ make clean && make && make install
```

The unit tests can be built with the same option: `make -C test/utest USE_AVX512=1` .

GNSS SDR Utilities and APs

Pocket SDR includes the following utilities for the Pocket SDR FE.

- **pocket_scan**: Scans and lists USB Devices.
- **pocket_conf**: Configures Pocket SDR FE device.
- **pocket_dump**: Captures and dumps IF data from the Pocket SDR FE device

Pocket SDR also provides the following GNSS SDR APs:

- **pocket_psd.py** : Plots PSD and histograms of IF data.
- **pocket_acq.py** : Performs GNSS signal acquisition from IF data.
- **pocket_trk.py** : Tracks GNSS signals and decodes navigation data in IF data.
- **pocket_snap.py**: Executes snapshot positioning using captured IF data.
- **pocket_sdr.py** : A GUI-based GNSS SDR receiver application.
- **pocket_plot.py**: Plots receiver logs generated by pocket_trk or pocket_sdr.py.
- **pocket_acq** : A C-version of pocket_acq.py (w/o graphical plots).
- **pocket_trk** : A C-version of pocket_trk.py (w/o graphical plots).
- **pocket_snap** : A C-version of pocket_snap.py.
- **pocket_web** : A GNSS SDR receiver server controlled from a Web UI.

For more details about these utilities and APs, please refer to **Pocket SDR Command References**.

GUI-based Real-Time GNSS SDR Receiver AP

Starting from version 0.13, Pocket SDR includes a GUI-based real-time GNSS SDR receiver AP, **pocket_sdr.py**. To execute the application, follow these steps:

```
$ chmod +x <install_dir>/python/pocket_sdr.py
$ sudo ./<install_dir>/python/pocket_sdr.py
or alternatively:
$ sudo python <install_dir>/python/pocket_sdr.py
```

For more information about this application, please refer to **pocket_sdr.py help**.

Web UI-based Real-Time GNSS SDR Receiver AP

Starting from version 0.20, Pocket SDR includes **pocket_web**, a real-time GNSS SDR receiver AP controlled from a Web UI in a browser. It provides the same receiver as pocket_trk and the same screens as pocket_sdr.py, but needs no GUI toolkit on the machine running the receiver, so it also suits a headless Raspberry Pi. To execute the AP:

```
$ cd <install_dir>/bin
$ sudo ./pocket_web -web 8080
```

Then open `http://localhost:8080/` in a browser. The receiver starts stopped: set the input source, output streams and signals in the Input, Output and Signal Options pages, and press Start. The settings are saved when the receiver is stopped and when the AP exits, and are restored at the next start, so `-start` starts the receiver right away with them.

To reach the AP from another PC, a tablet or a phone, bind it to all interfaces:

```
$ sudo ./pocket_web -web 0.0.0.0:8080
```

Note that the interface is unauthenticated and unencrypted. Use it in a trusted LAN only, or front it with a reverse proxy providing HTTPS and authentication.

`run_sdr.sh` in the top directory starts and stops the server:

```
$ ./run_sdr.sh          # start
$ ./run_sdr.sh -stop    # stop
```

Always stop the AP this way or with Ctrl-C. If the process is killed, the RF frontend is left streaming and needs a USB reset to recover (pocket_web recovers it at the next start, but the running capture is lost).

For more information about this AP, please refer to **Pocket SDR Command References**.

Execution Examples of Utilities and APs

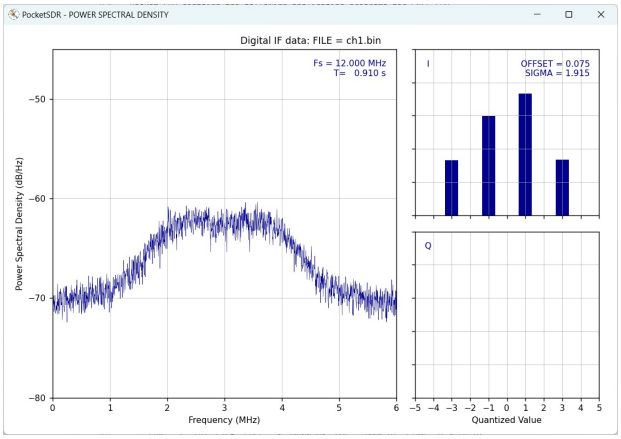
Here are some examples of how to execute the utilities and the APs:

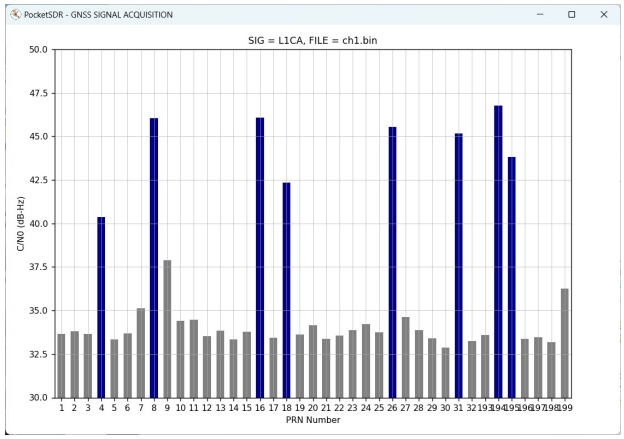
```
$ sudo pocket_conf
...
$ sudo pocket_conf conf/pocket_L1L6_12MHz.conf
Pocket SDR device settings are changed.

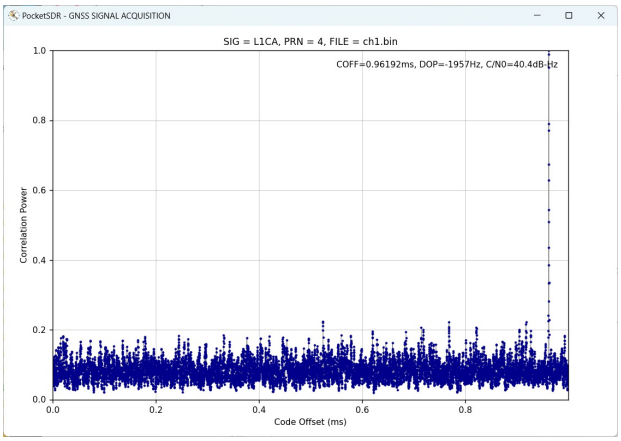
$ sudo pocket_dump -t 5 ch1.bin ch2.bin
  TIME(s)    T   CH1(Bytes)   T   CH2(Bytes)   RATE(Ks/s)
      5.0    I     60047360  IQ    120094720      11985.5

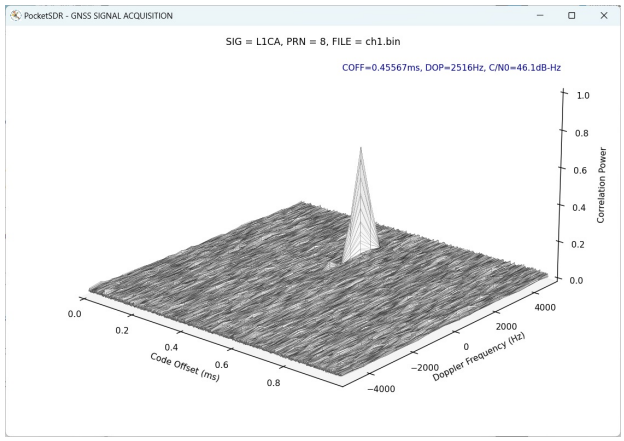
$ pocket_psd.py ch1.bin -f 12 -h

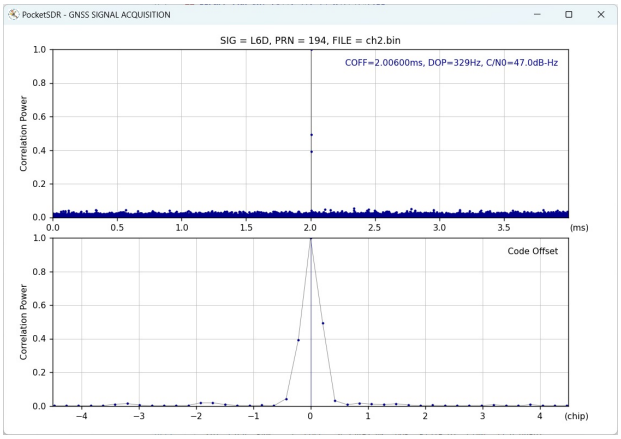
$ pocket_acq.py ch1.bin -f 12 -sig L1CA -prn 1-32,193-199
SIG= L1CA, PRN=   1, COFF=  0.23492 ms, DOP= -1519 Hz, C/N0= 33.6 dB-Hz
SIG= L1CA, PRN=   2, COFF=  0.98558 ms, DOP=  2528 Hz, C/N0= 33.8 dB-Hz
SIG= L1CA, PRN=   3, COFF=  0.96792 ms, DOP=  3901 Hz, C/N0= 33.7 dB-Hz
SIG= L1CA, PRN=   4, COFF=  0.96192 ms, DOP= -1957 Hz, C/N0= 40.4 dB-Hz
...
$ pocket_acq.py ch1.bin -f 12 -sig L1CA -prn 4
...
$ pocket_acq.py ch1.bin -f 12 -sig L1CA -prn 8 -3d
...
$ pocket_acq.py ch2.bin -f 12 -sig L6D -prn 194 -p
...
$ pocket_trk.py ch1.bin -f 12 -sig L1CA -prn 1-32
...
$ pocket_trk.py ch1.bin -f 12 -sig E1B -prn 18 -p
...
$ pocket_trk.py ch2.bin -f 12 -sig E6B -prn 4 -log trk.log -p -ts 0.2
...
```

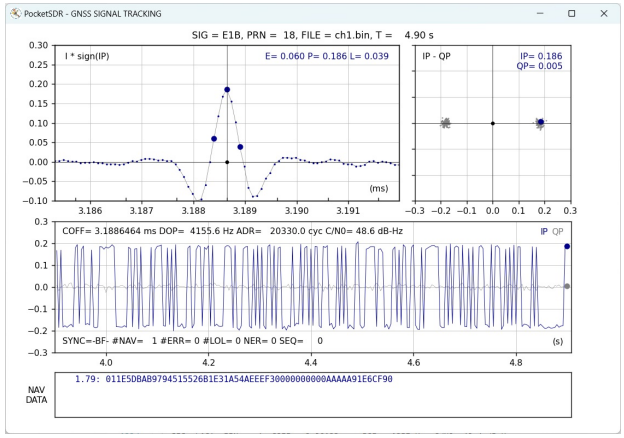

PocketSDR - POWER SPECTRAL DENSITY
Digital IF data: FILE = ch1.bin
Fs = 12.000 MHz
T= 0.910 s
I
OFFSET = 0.075
SIGMA = 1.915
Q
Power Spectral Density (dB/Hz)
Frequency (MHz)
Quantized Value

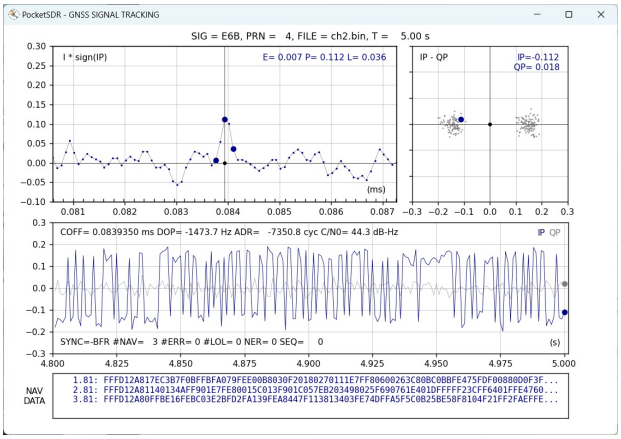

PocketSDR - GNSS SIGNAL ACQUISITION
SIG = L1CA, FILE = ch1.bin
C/N0 (dB-Hz)
PRN Number


PocketSDR - GNSS SIGNAL ACQUISITION
SIG = L1CA, PRN = 4, FILE = ch1.bin
COFF=0.96192ms, DOP=-1957Hz, C/N0=40.4dB-Hz
Correlation Power
Code Offset (ms)


PocketSDR - GNSS SIGNAL ACQUISITION
SIG = L1CA, PRN = 8, FILE = ch1.bin
COFF=0.45567ms, DOP=2516Hz, C/N0=46.1dB-Hz
Correlation Power
Code Offset (ms)
Doppler Frequency (Hz)


PocketSDR - GNSS SIGNAL ACQUISITION
SIG = L6D, PRN = 194, FILE = ch2.bin
COFF=2.00600ms, DOP=329Hz, C/N0=47.0dB-Hz
Correlation Power
(ms)
Code Offset
(chip)


PocketSDR - GNSS SIGNAL TRACKING
SIG = E1B, PRN = 18, FILE = ch1.bin, T = 4.90 s
I * sign(IP)
E= 0.060 P= 0.186 L= 0.039
IP - QP
IP= 0.186
QP= 0.005
(ms)
COFF= 3.1886464 ms DOP= 4155.6 Hz ADR= 20330.0 cyc C/N0= 48.6 dB-Hz
IP QP
SYNC=-BF- #NAV= 1 #ERR= 0 #LOL= 0 NER= 0 SEQ= 0
(s)
NAV DATA
1.79: 011E5DBAB97945155268E31A54AEEEF3000000000AAAAA91E6CF90


PocketSDR - GNSS SIGNAL TRACKING
SIG = E6B, PRN = 4, FILE = ch2.bin, T = 5.00 s
I * sign(IP)
E= 0.007 P= 0.112 L= 0.036
IP - QP
IP=-0.112
QP= 0.018
(ms)
COFF= 0.0839350 ms DOP= -1473.7 Hz ADR= -7350.8 cyc C/N0= 44.3 dB-Hz
IP QP
SYNC=-BFR #NAV= 3 #ERR= 0 #LOL= 0 NER= 0 SEQ= 0
(s)
NAV DATA
1.81: FFFD12A817EC3B7F0BFFBFA079FEE00B8030F20180270111E7FF80600263C80BC0BBFE475FDF00880D0F3F...
2.81: FFFD12A81140134AFF901E7FE80015C013F901C057EB203498025F690761E401DFFFF23CFF6401FFE4760...
3.81: FFFD12A80FFBE16FEBC03E2BFD2FA139FEA8447F113813403FE74DFFA5F5C0B25BE58F8104F21FF2FAEFFE...

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Pocket SDR is released for **research and educational use**. Public availability of the source code, hardware design data, and firmware in this repository does not exempt users from compliance with applicable export control regulations, including the Japan Foreign Exchange and Foreign Trade Act (外為法), the US Export Administration Regulations (EAR), the EU dual-use regulation (EU 2021/821), and the Wassenaar Arrangement (Category 7 ? Navigation and Avionics). The 8-channel hardware (FE_8CH) combined with the antenna array calibration and digital beam-forming features added in v0.15 may be regarded as dual-use technology under some jurisdictions; users should perform their own classification (該非判定) before cross-border transfer.

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References

[1] *NMEA 0183: Standard for Interfacing Marine Electronic Devices*, National Marine Electronics Association and International Marine Electronics Association, 2013.

[2] *RTCM 10403.4 with Amendment 1: Differential GNSS (Global Navigation Satellite Systems) Service - Version 3*, Radio Technical Commission for Maritime Services, November 1, 2024

History

- **2021-10-20 (v0.1)**: Initial draft version.
- **2021-10-25 (v0.2)**: Added rebuild firmware and write firmware image to Pocket SDR.
- **2021-12-01 (v0.3)**: Added and modified Python scripts.
- **2021-12-25 (v0.4)**: Added and modified Python scripts.
- **2022-01-05 (v0.5)**: Fixed several issues.
- **2022-01-13 (v0.6)**: Added and modified Python scripts.
- **2022-02-15 (v0.7)**: Improved performance and added more Python scripts.
- **2022-07-08 (v0.8)**: Added C-version of pocket_acq.py and pocket_trk.py.
- **2024-01-03 (v0.9)**: Added C-version of pocket_snap.py. pocket_trk now supports multi-signal and multi-threading.
- **2024-01-12 (v0.10)**: Added support for NavIC L1-SPS-D, L1-SPS-P, GLONASS L1OCd, L1OCp, and L2OCp.
- **2024-01-25 (v0.11)**: Added support for decoding GLONASS L1OCd NAV data and NB-LDPC error correction for BDS B1C, B2a, and B2b.
- **2024-05-28 (v0.12)**: Performance optimizations. Added support for PVT generation, RTCM3, and NMEA outputs.
- **2024-07-04 (v0.13)**: Added GUI-based GNSS SDR receiver AP. Added support for macOS and Raspberry Pi OS.
- **2025-03-21 (v0.14)**: Added Pocket SDR FE 8CH. Fixed various issues.
- **2026-05-12 (v0.15)**: Added antenna array calibration (per-epoch EKF on single-difference carrier-phase residuals) and digital beam-forming. Added LUT-based array combine (~2x throughput at narch=8). Added GUI Array tab with calibration / beam control and gain pattern overlay. Added Galileo E5ABQ signal support. Added SoapySDR / LimeSDR support on Windows (UCRT64 + radioconda). Refactored array calibration / beam-forming API.
- **2026-06-01 (v0.16)**: Added fast acquisition using saved ephemeris / almanac data, latest FIX position, and receiver clock drift rate. Added `.pocket_navdata.csv` persistence for almanacs and last FIX state with timestamp validation. Added SoapySDR input support to `pocket_dump`, CS8 / CS16 support to `pocket_acq`, and helper scripts for SoapySDR capture / tracking. Added `USE_SOAPY` and `USE_FFTW` make options. Fixed FE 2CH I-sampling real-to-complex conversion and updated command/API references.
- **2026-06-15 (v0.17)**: Improved signal-loss decision with a carrier lock detector (PLI) and pessimistic counter in addition to C/N0 (`thres_pli`, `lost_th` options). Replaced external Viterbi / Reed-Solomon / LDPC decoders with internal implementations. Improved the symbol-sync algorithm. Improved the `pocket_sdr.py` UI (options Load / Save, etc.).
- **2026-07-06 (v0.18)**: Reworked the QZSS L6 CSK decoder: chip-domain FFT symbol detection with sampling-rate-independent cost, E/P/L correlations by the standard correlator with the CSK-shifted code (improves L6 pseudorange quality), and a narrowed peak search after frame sync. L6 tracking is 3-8x faster; 8 L6 channels at 24 Msps fit on a Raspberry Pi 5 (see doc/update_csk_decoder.md). Fused carrier wipeoff into the standard and FFT correlators and unified the int8 code banks. Optimized NEON paths for Raspberry Pi 5 (SDOT). Added option `-w` for FFTW wisdom import.

- **2026-07-24 (v0.19)**: Added int8 code support to the correlators for multi-level sub-carrier replicas: CBOC for Galileo E1B / E1C and AltBOC for E5. Improved E5 AltBOC tracking with pre-combined complex replicas on a restored complex correlator path. Added extended coherent tracking for pilot signals and low-C/N0 acquisition / tracking options (`t_acq_ext` , `t_coh` , `thres_cn0_ext`). Removed the C/N0 estimation bias at low C/N0. Widened the pseudorange lower bound for GEO / IGSO epoch seeding. Generalized the output streams: up to 8 streams with selectable types (NMEA, RTCM3, receiver log, IF data log), e.g. NMEA to a file and a TCP server simultaneously (`-nmea` / `-rtcm` / `-log` / `-raw` repeatable in pocket_trk; new Output Options UI in pocket_sdr.py).
- **2026-08-08 (v0.20)**: Added a Web UI. The new AP **pocket_web** embeds an HTTP + WebSocket server (src/sdr_web.c) and serves a static front end (html/, plain JS, no dependency or build step), so a browser drives the whole receiver, even on a headless Raspberry Pi, with the settings kept across sessions (see doc/design_web_ui.md). Recovered a Pocket SDR FE left streaming by a killed session. pocket_trk keeps its command line.
- **2026-10-21 (v0.21)**: Added multi-device and multi-channel SoapySDR input: devices sharing a common 10 MHz and 1PPS are opened as one multi-channel frontend (up to 8 RF channels) with the streams started at a common time, including `-RFCH` assignment and `-arch` array combining. Added SoapySDR options `-EXT_CLOCK` and `-gpio` , dropped-sample zero-fill, sensor logging (`-sens`), and multi-channel SoapySDR capture by pocket_dump for replay. Enlarged the Pocket SDR FE raw data buffer to 128 MB with overflow logging and a raised receiver thread priority, fixing the loss of all signals at once at high data rates. Fast acquisition now predicts the satellites at the receiver time instead of the host PC clock, so it also works for a GNSS simulator and replayed IF data. Added AVX512 kernels for the carrier mixer, correlators, LPF and complex multiply (`USE_AVX512=1`). Added a dark theme to pocket_sdr.py. Fixed the satellite week not updated on the TOW rollover, sat_id() for BeiDou B1C / B2a / B2I, the start-time dependent fractional carrier phase bias between channels that broke the integer double-difference ambiguities (GitHub issue #97), and potential buffer overflows in the pocket_web JSON replies (GitHub issue #98).
